B. Tech Degree VII Semester (Supplementary) Examination June 2011

EB/EC/EI 705 (D) MECHATRONICS

(2006 Scheme)

Time: 3 Hours | Maximum Marks: 100

PART – A

(Answer *all* questions)

(8 x 5 = 40)

I. (a) Explain the working of pyro electric sensors.
(b) Write short notes on static and dynamic characteristics.
(c) With necessary diagram explain hydraulic power supply.
(d) Explain how stepper motors are selected for specific applications.
(e) Explain the input output relation of the rack and pinion.
(f) Draw and explain the block diagram representation of the DC motor circuits.
(g) Write short notes on FMS.
(h) What is watch dog timer? Write a PLC program for the watch dog timer.

PART – B

(4 x 15 = 60)

II. What is the need for signal conditioning? Explain any four signal conditioning techniques used in mechatronic systems. (15)

OR

III. (a) Design a temperature alarm system in which the alarm is set when the temperature of a liquid rises above 40°C. The liquid is normally at 30°C. The output from the system must be a W signal to operate the alarm (Choose nickel as the resistance element having a temperature coefficient of resistance as 0.0067/K). (5)
(b) What are data acquisition systems? With necessary block diagrams explain DAS. (10)

IV. Explain the following:
a) Kinematic chains
b) Serve motors
c) Rotary actuators (15)

OR

V. (a) Explain the different types of Pressure Control Valves. (10)
(b) With necessary diagrams explain the functioning of four-bar chain. (5)

VI. Define MEMS. What are the various manufacturing processes of MEMS? List its advantages and applications. (15)

OR

VII. Obtain the relationship for the basic building blocks of fluid flow systems. (15)

VIII. (a) Explain the architecture of a PLC with the help of a block diagram (8)
(b) With a neat block diagram explain a CNC system. (7)

OR

IX. (a) With the help of block diagram explain Digital Controllers. (8)
(b) What is derivative control? Why derivative control is always combined with proportional control? Explain how PD control is implemented with OP-AMP circuits. (7)

***